

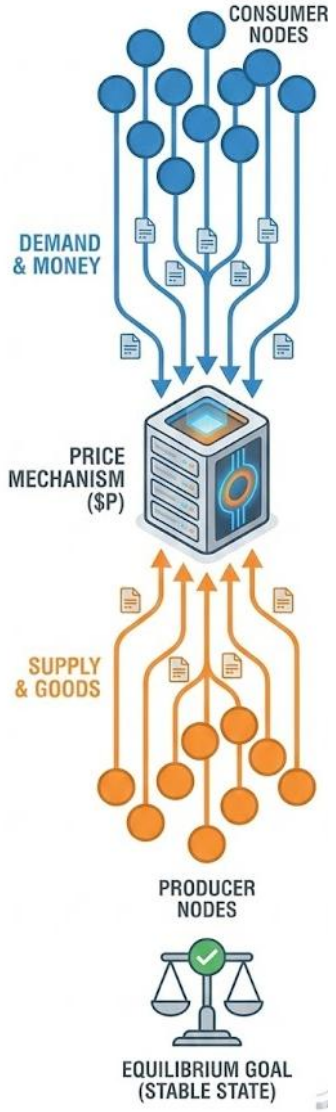
Market Dynamics: Equilibrium, Floating Prices, and the Oil Economy

ITGE275 Economics

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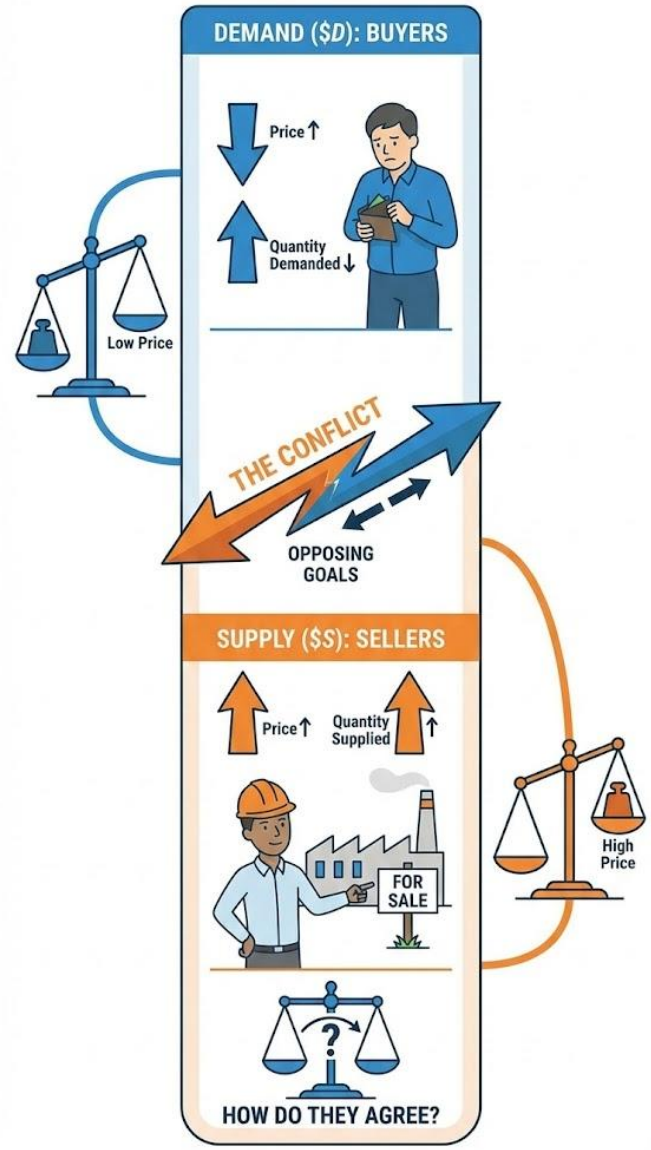
The Market as a Distributed System

- **Concept:** A market is not just a place; it is a decentralized computing system.
- **Components:**
  - **Nodes (Agents):** Millions of Consumers and Producers acting independently.
  - **Signals:** Price (***P***) is the signal that coordinates their behavior.
  - **Protocol:** The Law of Supply and Demand.
- **The Goal:** To find a state where the system is stable and cleared (Equilibrium).



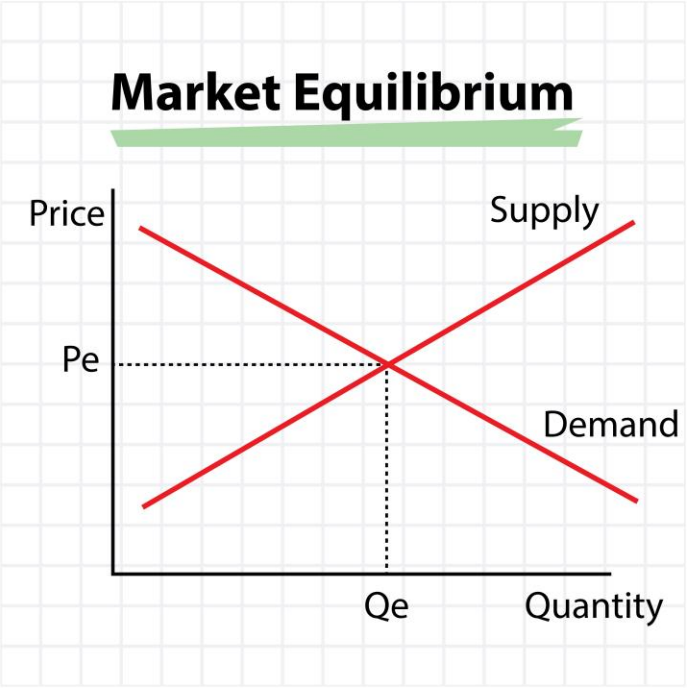
Demand and Supply Review

- **Demand (*D*):** The behavior of buyers.
  - *Law:* Price ↑, Quantity Demanded ↓ (Inverse relationship).
- **Supply (*S*):** The behavior of sellers.
  - *Law:* Price ↑, Quantity Supplied ↑ (Direct relationship).
- **The Conflict:** Buyers want low prices; Sellers want high prices. How do they agree?



Market Equilibrium

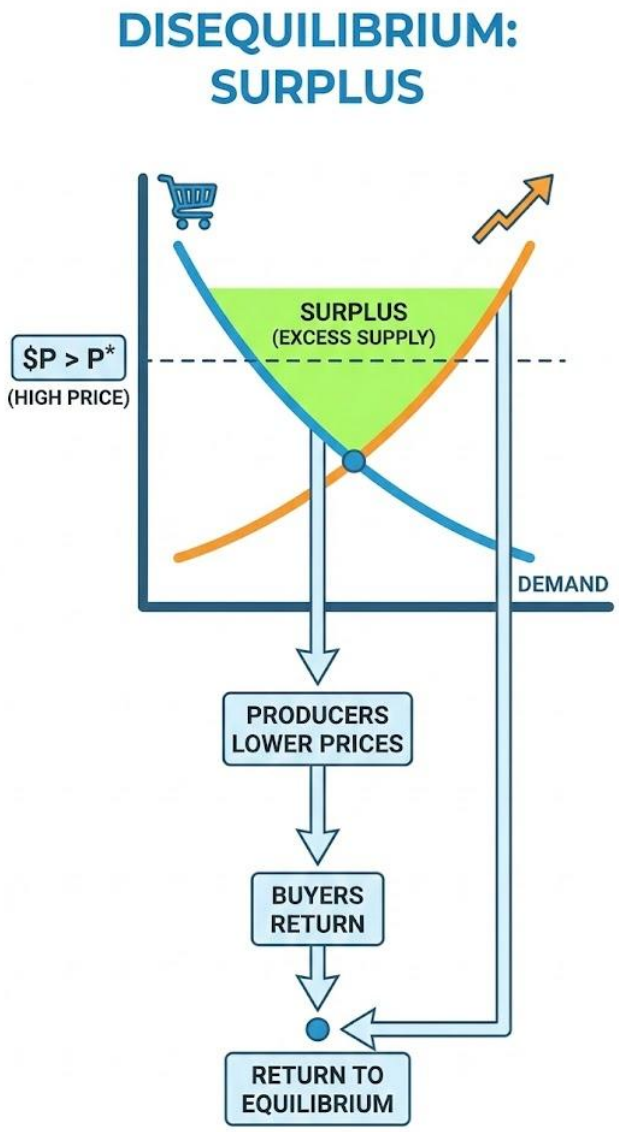
- **Definition:** The specific price point where the Quantity Demanded exactly equals the Quantity Supplied ($Q_d = Q_s$).
- **The "Clearing" Price:** At this price (P^*), there is:
  - **No Surplus:** No unsold goods piling up in warehouses.
  - **No Shortage:** No angry lines of customers empty-handed.
- **Stability:** Once reached, there is no tendency for the price to change unless a shock occurs.





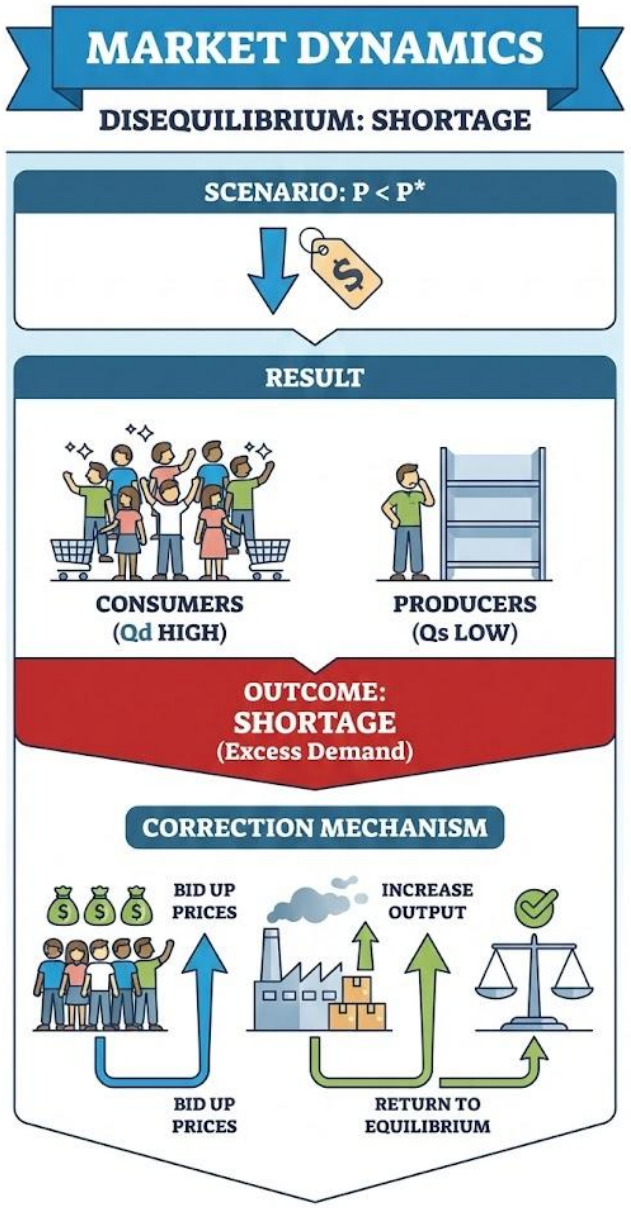
Disequilibrium: Surplus

- **Scenario:** What if the Price is *above* Equilibrium ($P > P^*$)?
- **Result:**
  - Producers want to sell A LOT (Q_s is high).
  - Consumers want to buy VERY LITTLE (Q_d is low).
- **Outcome:** A **Surplus** (Excess Supply).
- **Correction Mechanism:** Producers see unsold inventory → They lower prices → Buyers return → Return to Equilibrium.



Disequilibrium: Shortage

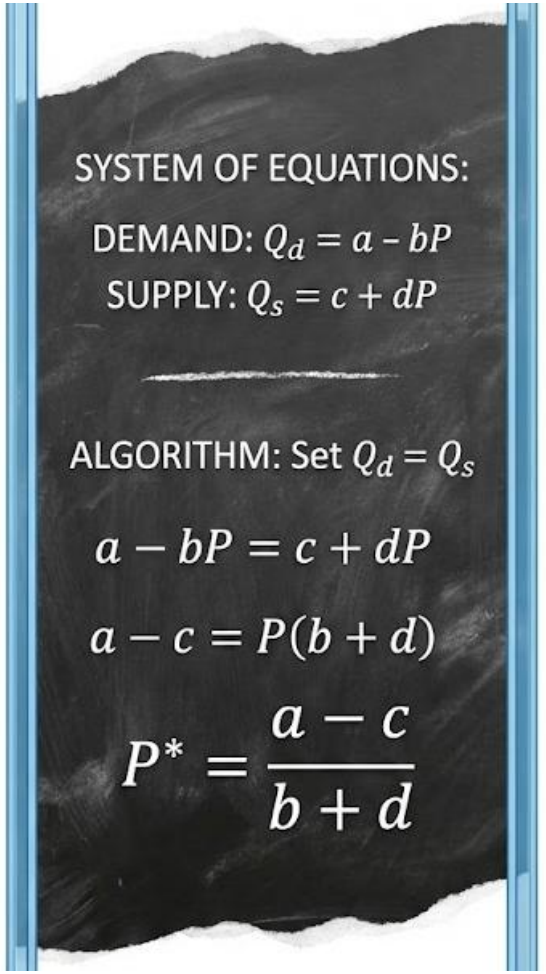
- **Scenario:** What if the Price is *below* Equilibrium ($P < P^*$)?
- **Result:**
  - Consumers want to buy A LOT (Q_d is high).
  - Producers want to sell VERY LITTLE (Q_s is low).
- **Outcome:** A **Shortage** (Excess Demand).
- **Correction Mechanism:** Consumers bid up prices → Producers increase output → Return to Equilibrium.



The Mathematics of Equilibrium

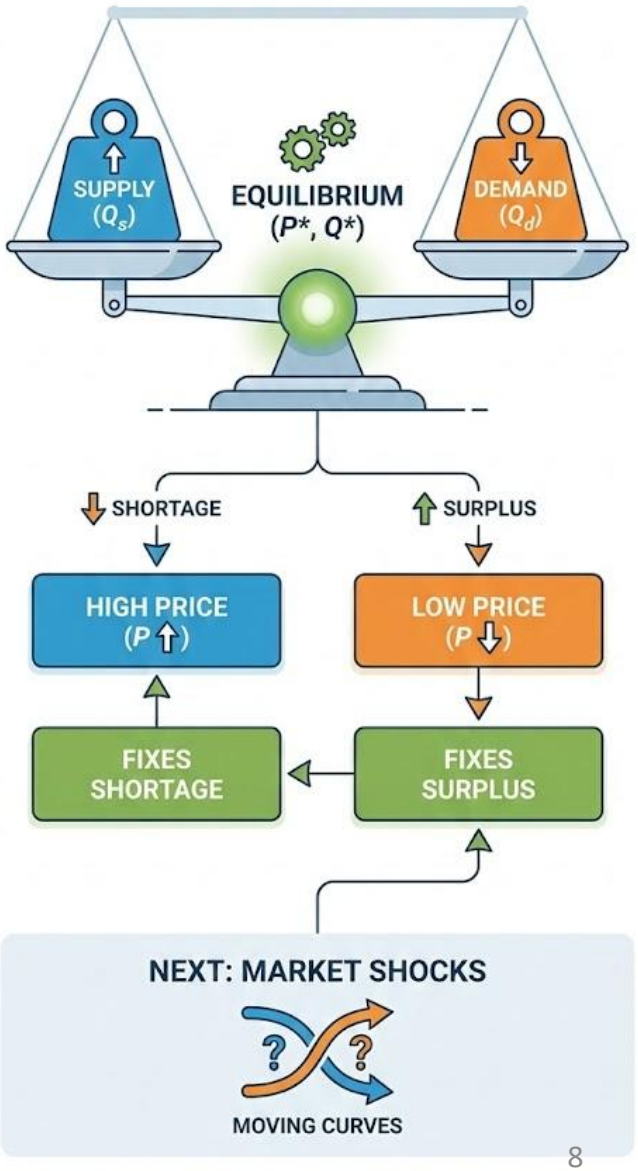
- **System of Equations:**
  - Demand: $Q_d = a - bP$
  - Supply: $Q_s = c + dP$
- **Algorithm:** Set $Q_d = Q_s$ and solve for P.
  - $a - bP = c + dP$
  - $a - c = P(b + d)$

  ➡ $P^* = \frac{a - c}{b + d}$
- **In the Lab:** We will implement this as a Python function `solve_equilibrium(a, b, c, d)`.



Part 1 Summary

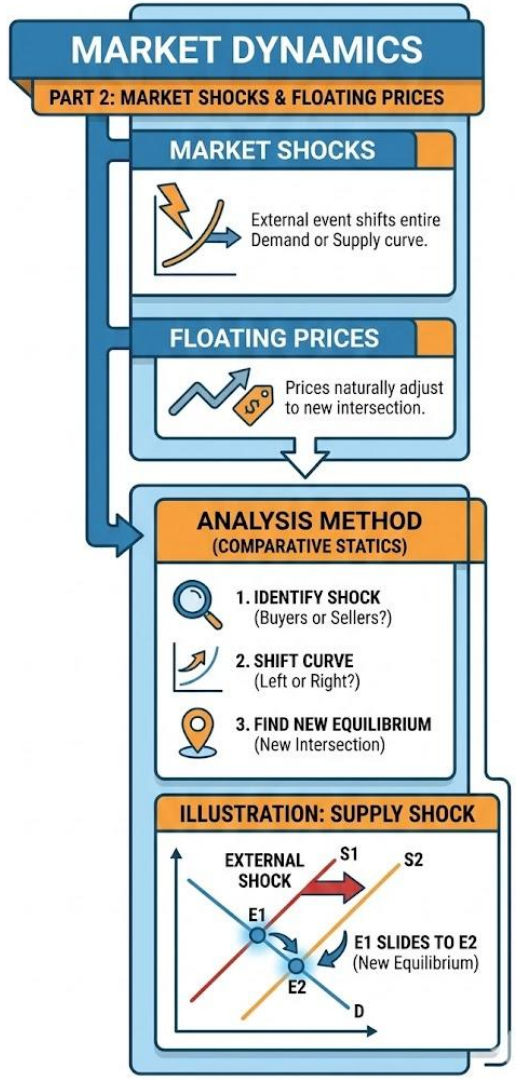
- Markets are self-correcting systems.
- **Equilibrium (P^*, Q^*):** The point of balance where $Q_d = Q_s$.
- **Price Signal:** High prices fix shortages; Low prices fix surpluses.
- **Next:** What happens when the curves move? (Shocks).



Part 2: Market Shocks & Floating Prices

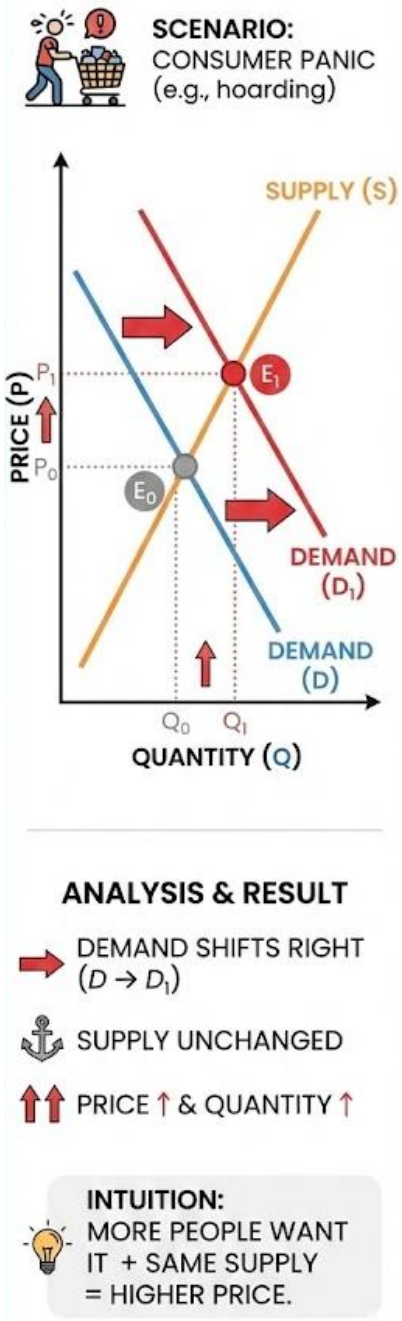
Comparative Statics (Analyzing Shocks)

- The Equilibrium point is not permanent. It moves when the underlying curves shift.
- **Shock:** An external event that shifts the entire Demand or Supply curve.
- **Floating Prices:** In a free market, prices naturally adjust ("float") to the new intersection point.
- **Analysis Method:**
  - Identify the Shock (Who is affected? Buyers or Sellers?).
  - Shift the Curve (Left or Right?).
  - Find the New Equilibrium.



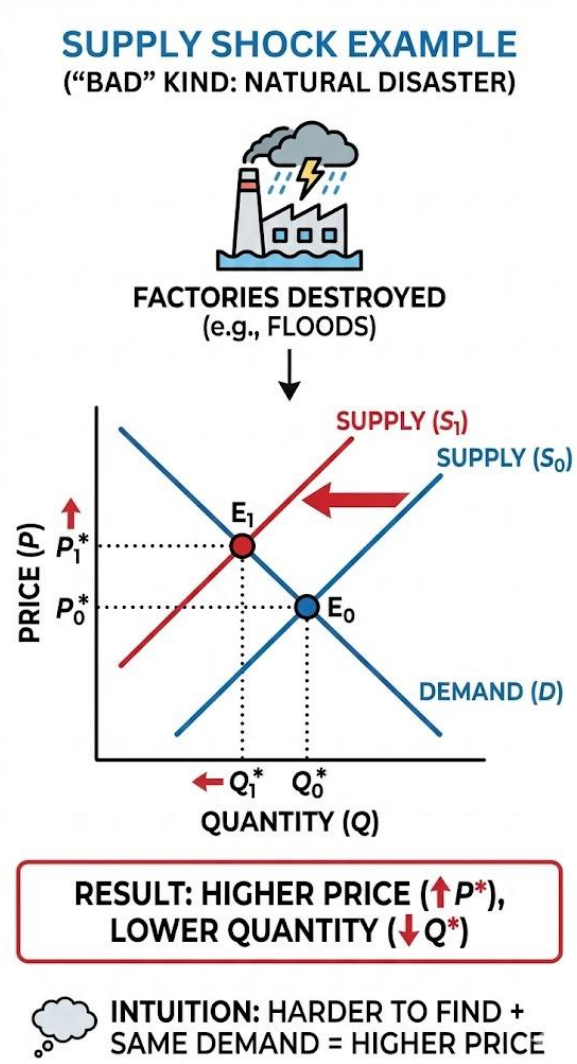
Demand Shock Example

- **Scenario:** "Consumer Panic" (e.g., hoarding toilet paper during a pandemic).
- **Analysis:**
  - **Curve:** Demand shifts Right ($D \rightarrow D_1$).
  - **Supply:** Unchanged (Producers haven't built new factories yet).
- **Result:** Equilibrium Price ↑, Equilibrium Quantity ↑.
- **Intuition:** More people want it + Same supply = Higher Price.



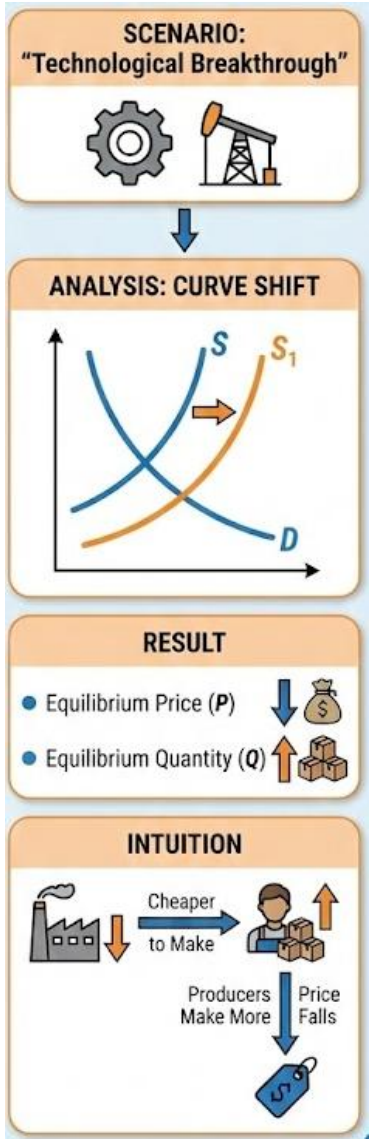
Supply Shock Example (The "Bad" Kind)

- **Scenario:** "Natural Disaster" destroys factories (e.g., Floods in Thailand hard drive factories).
- **Analysis:**
  - **Curve:** Supply shifts Left ($S \rightarrow S_1$).
  - **Demand:** Unchanged.
- **Result:** Equilibrium Price ↑, Equilibrium Quantity ↓.
- **Intuition:** Stuff is harder to find + Same demand = Higher Price.



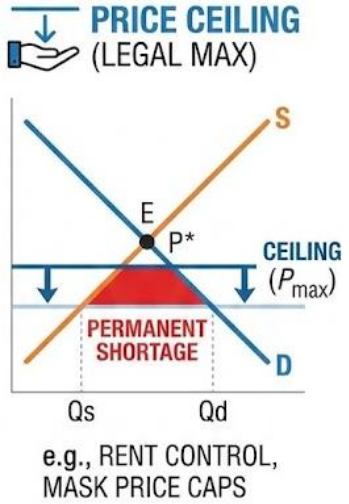
Supply Shock Example (The "Good" Kind)

- **Scenario:** "Technological Breakthrough" (e.g., Fracking technology for oil).
- **Analysis:**
  - **Curve:** Supply shifts Right ($\rightarrow S_1$).
- **Result:** Equilibrium Price ↓, Equilibrium Quantity ↑.
- **Intuition:** It's cheaper to make ⟶ Producers make more ⟶ Price falls.

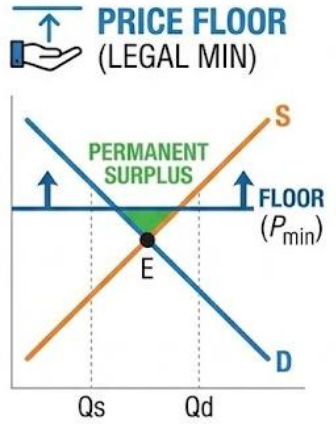


Price Controls (When Prices CAN'T Float)

- Sometimes governments intervene to stop floating prices.
- **Price Ceiling:** A legal maximum price (e.g., Rent Control, Mask price caps).
  - *Result:* If set below ***P****, it creates a **permanent Shortage**.
- **Price Floor:** A legal minimum price (e.g., Minimum Wage, Guaranteed Farm Prices).
  - *Result:* If set above ***P****, it creates a **permanent Surplus**.



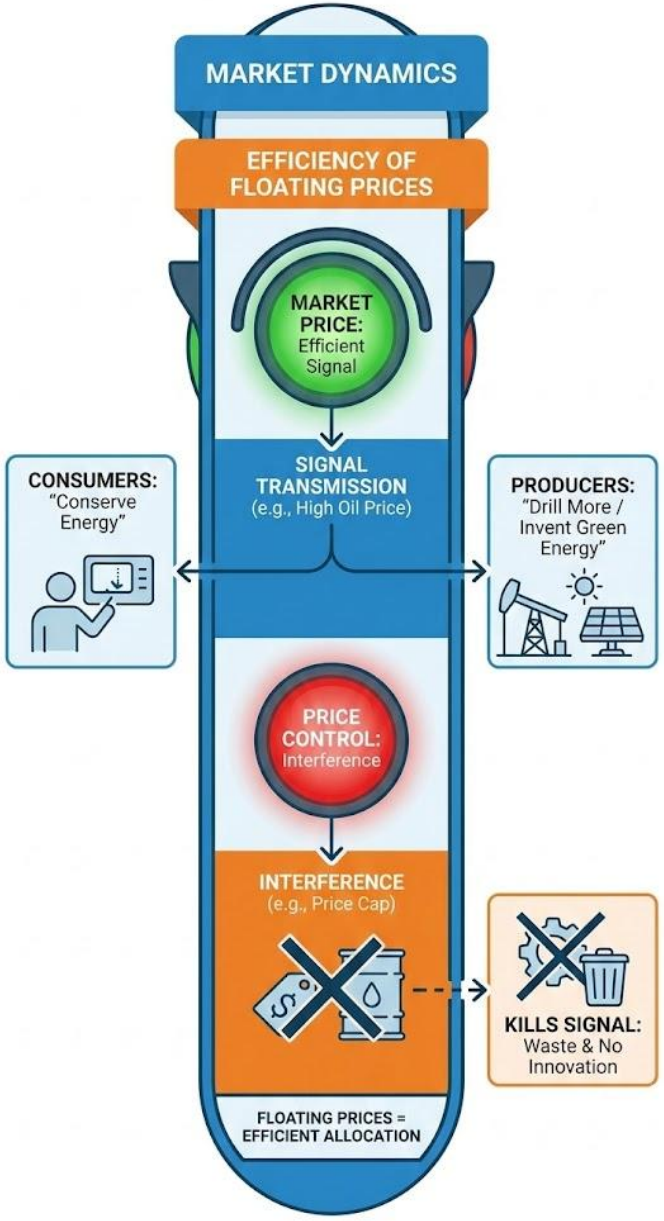
e.g., RENT CONTROL, MASK PRICE CAPS



e.g., MINIMUM WAGE, FARM PRICE SUPPORTS

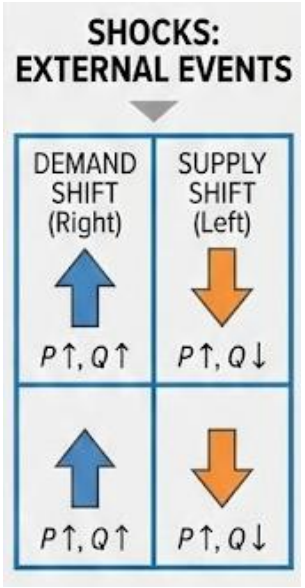
The Efficiency of Floating Prices

- **Why economists prefer floating prices:** They allocate resources efficiently without central planning.
- **Signal Transmission:**
  - High Oil Price → Signal to Consumers: "Conserve energy."
  - High Oil Price → Signal to Producers: "Drill more / Invent Green Energy."
- **Interference:** Capping the price kills the signal, leading to waste or lack of innovation.



Part 2 Summary

- **Shocks:** External events shift curves.
- **Demand Right:** P ↑, Q ↑
- **Supply Left:** P ↑, Q ↓
- **Price Controls:** Prevent equilibrium, causing shortages or surpluses.
- **Next:** Applying this to the most critical commodity—Oil.



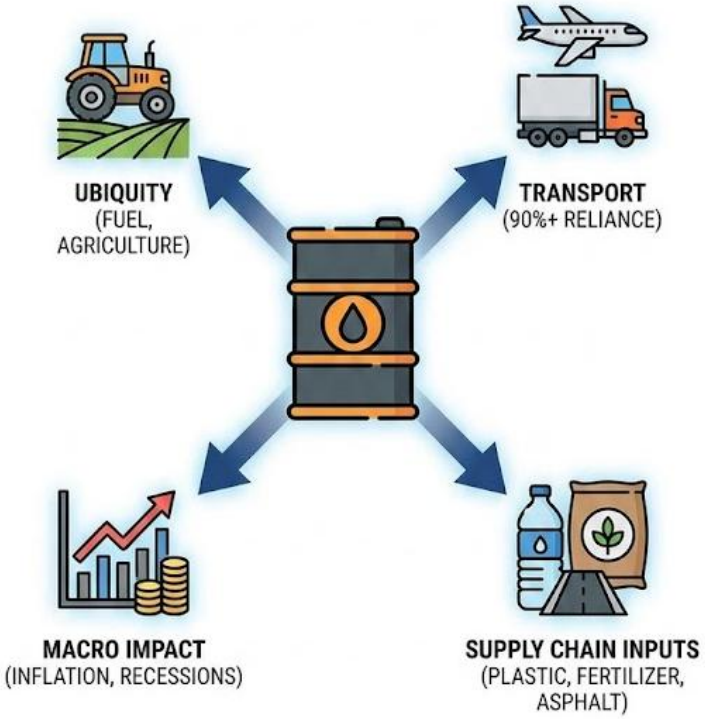




Part 3: The Oil Economy

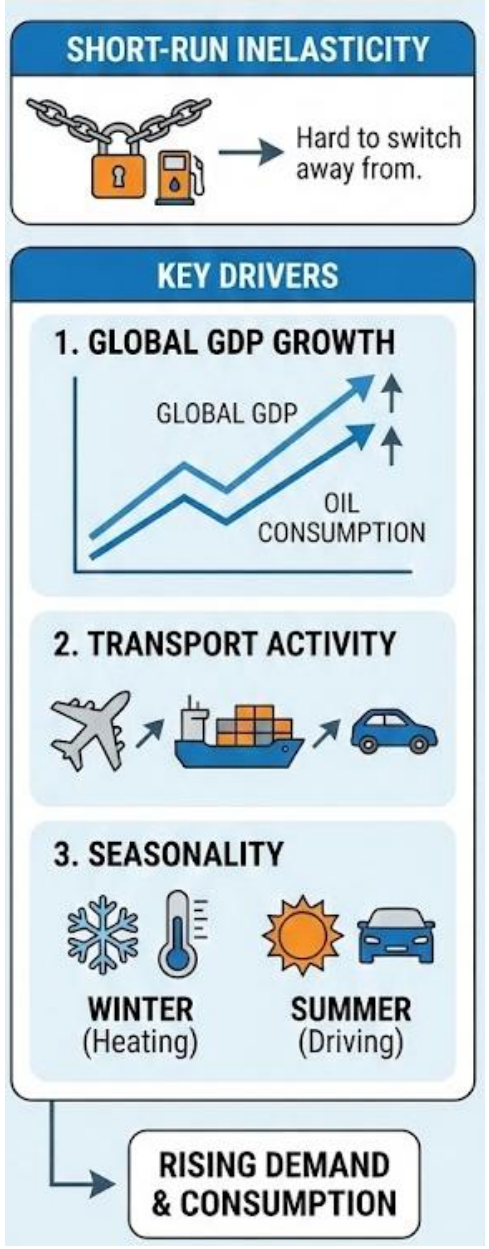
Oil - The Master Resource

- Why focus on Oil?
- ***Ubiquity***: It is an input in almost every supply chain (Plastic, Fertilizer, Fuel, Asphalt).
- **Transport:** 90%+ of global transport relies on oil derivatives.
- **Macro Impact:** Oil price volatility causes global inflation and recessions.



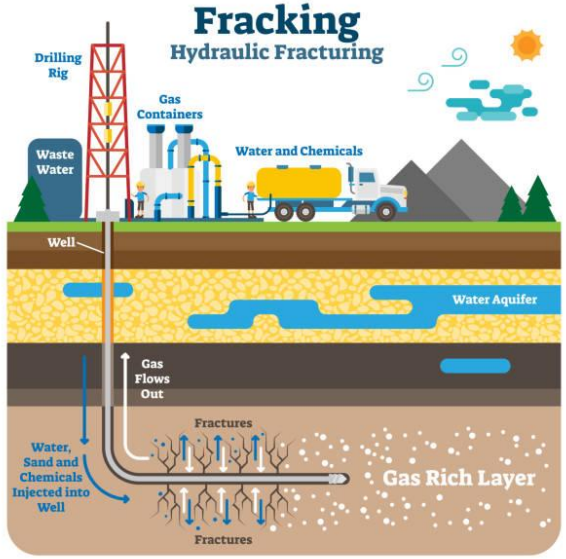
Determinants of Oil Demand

- Oil demand is **Inelastic** in the short run (hard to switch away from).
- **Drivers:**
  - **Global GDP Growth:** Growing economies (China, India) need more energy.
  - **Transport Activity:** Tourism, Shipping, Commuting.
  - **Seasonality:** Heating oil in winter, Driving in summer.

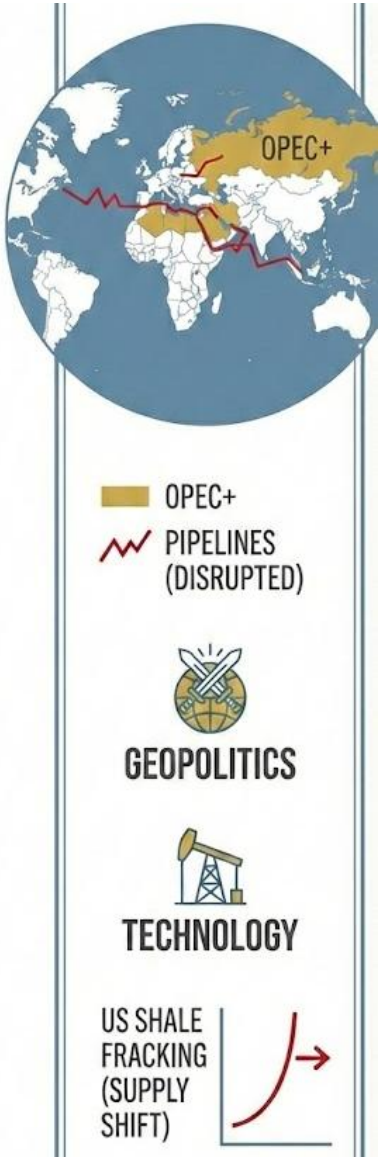


Determinants of Oil Supply

- **OPEC+:** A cartel of oil-producing nations that colludes to control supply and fix prices.
- **Geopolitics:** Wars in the Middle East or Russia disrupt supply lines.
- **Technology:** US Shale "Fracking" shifted the supply curve right in the 2010s.



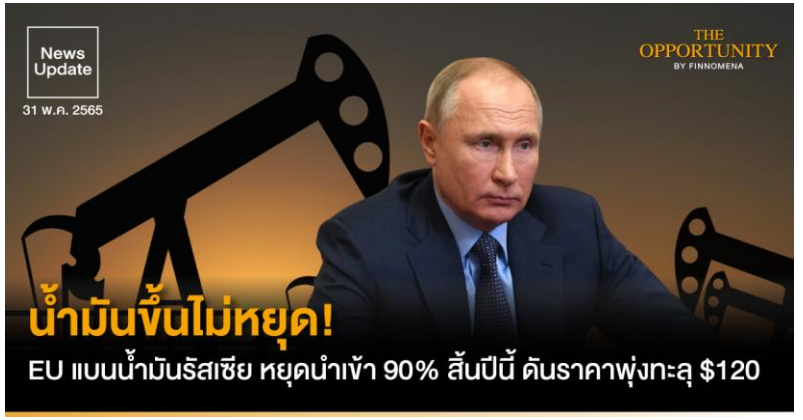
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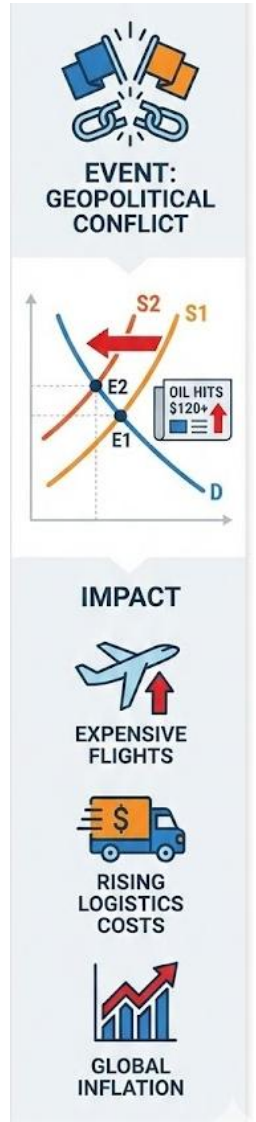


Case Study 1: The 2022 Supply Shock

- **Event:** Geopolitical conflict involving a major producer (Russia).
- **Market Analysis:**
  - Sanctions/Ban → **Supply Shift Left**.
  - Demand remained constant.
- **Result:** Prices spiked to >$100/barrel.
- **Impact:** Airline tickets became expensive, logistics costs rose, global inflation surged.

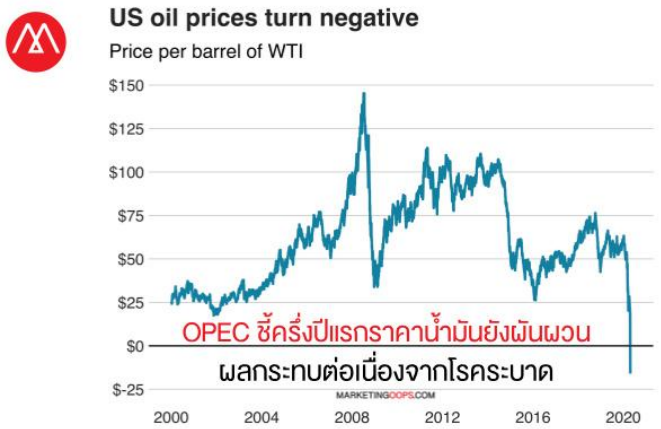






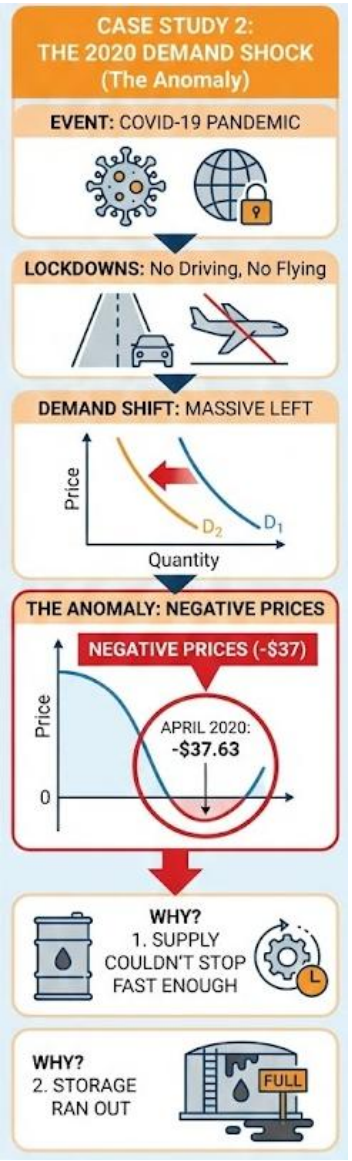
Case Study 2: The 2020 Demand Shock (The Anomaly)

- **Event:** COVID-19 Pandemic.
- **Market Analysis:**
  - Lockdowns ⟶ No driving, no flying.
  - **Demand Shift Massive Left**.
- **The Anomaly:** Prices turned **NEGATIVE** (-$37).
- **Why?** Supply couldn't stop fast enough, and **Storage** ran out. Producers paid buyers to take the oil because they had nowhere to put it. Keeping the oil costs $$$.



https://www.marketingoops.com/news/biz-news/opec-sees-oil-outlook-for-first-half-of-2021-full-of-risks/





The Local Pass-Through Effect

- How does global Brent Crude ($) affect Thai Diesel (THB)?
- **Pass-through:** The correlation is strong but rarely 1:1.
- **Buffers:**
  - **Exchange Rate:** THB/USD fluctuations.
  - **Government Subsidies:** The "Oil Fund" often caps diesel prices to prevent inflation (Price Ceiling).



https://www.tradingview.com/symbols/USDTHB/





Oil and the Stock Market

- **Inverse Correlation:**
  - High Oil Prices = High Costs for Airlines/Logistics.
  - Result: Airline stocks usually **drop** when Oil **rises**.
- **Direct Correlation:**
  - High Oil Prices = High Profits for Energy Companies (PTT, Exxon).
  - Result: Energy stocks **rise**.



Part 3 Summary

- Oil drives the global economy.
- It is subject to violent shocks (OPEC, War, Pandemics).
- Prices can go negative if storage is constrained.
- Local prices are buffered by policy.
- **Next:** Let's look at the data in Python.



Part 4: Tutorial & Homework Preview

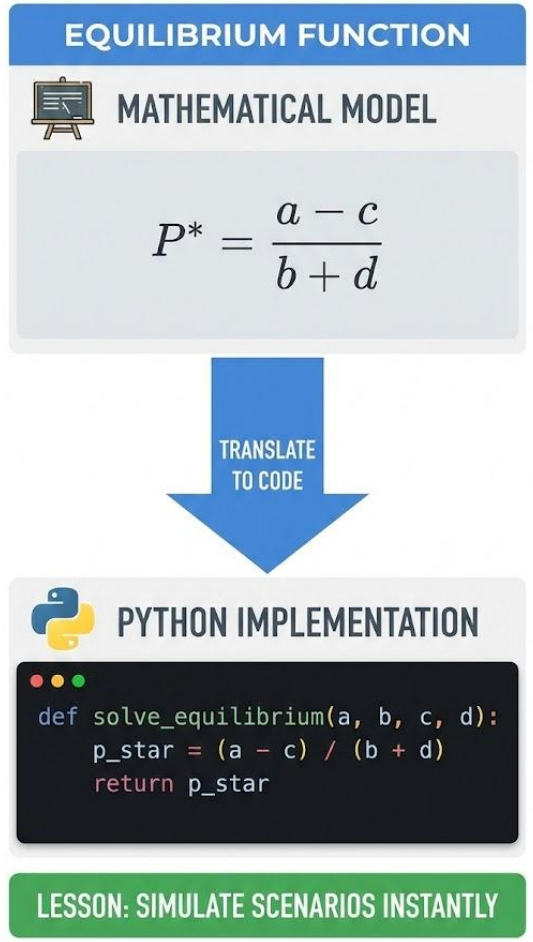
Tutorial Overview: Computational Market Dynamics

- **Goal:** Build a Market Simulator and Analyze Real Oil Data.
- **Tools:** pandas, matplotlib, seaborn.
- **Key Modules:**
  - **Equilibrium Solver:** Python function to find P^* and Q^*.
  - **Market Simulator:** Interactive widget to shock the economy.
  - **Oil Data Analysis:** Visualizing the 2020 crash and 2022 spike.



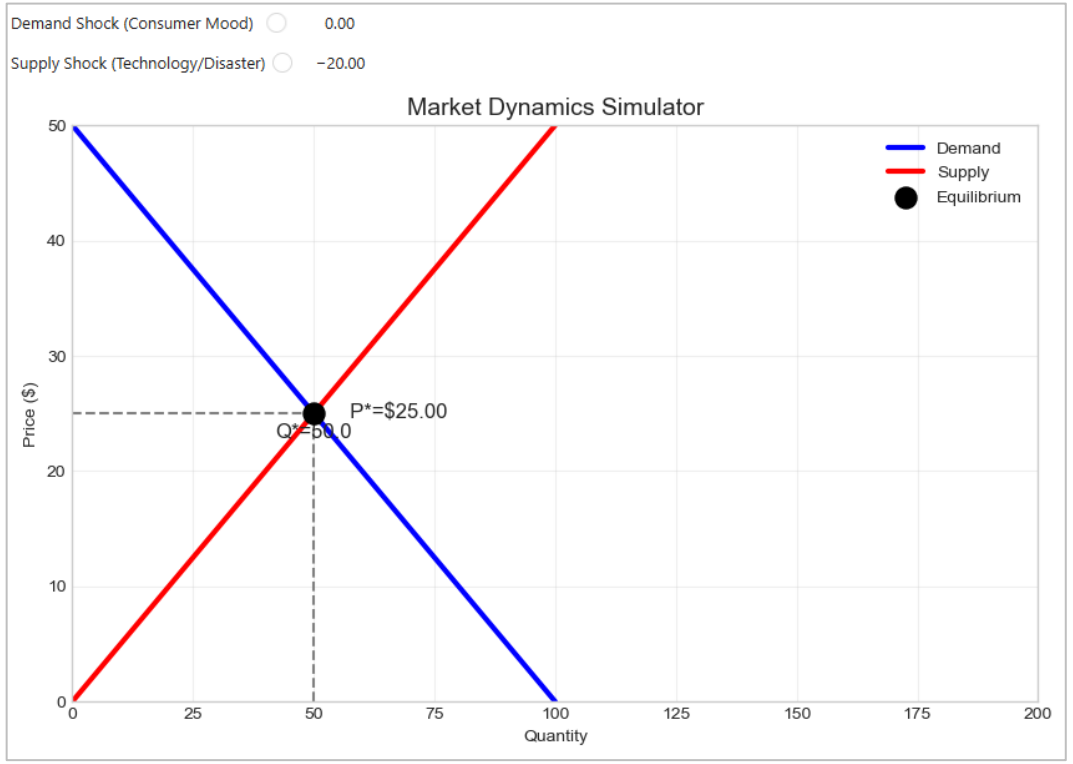
Building the Equilibrium Function

- We translate Math to Python:
  - Math: P^* = (a - c) / (b + d)
  - Python: p_star = (a - c) / (b + d)
- **Lesson:** Coding economic formulas allows us to simulate thousands of scenarios instantly.



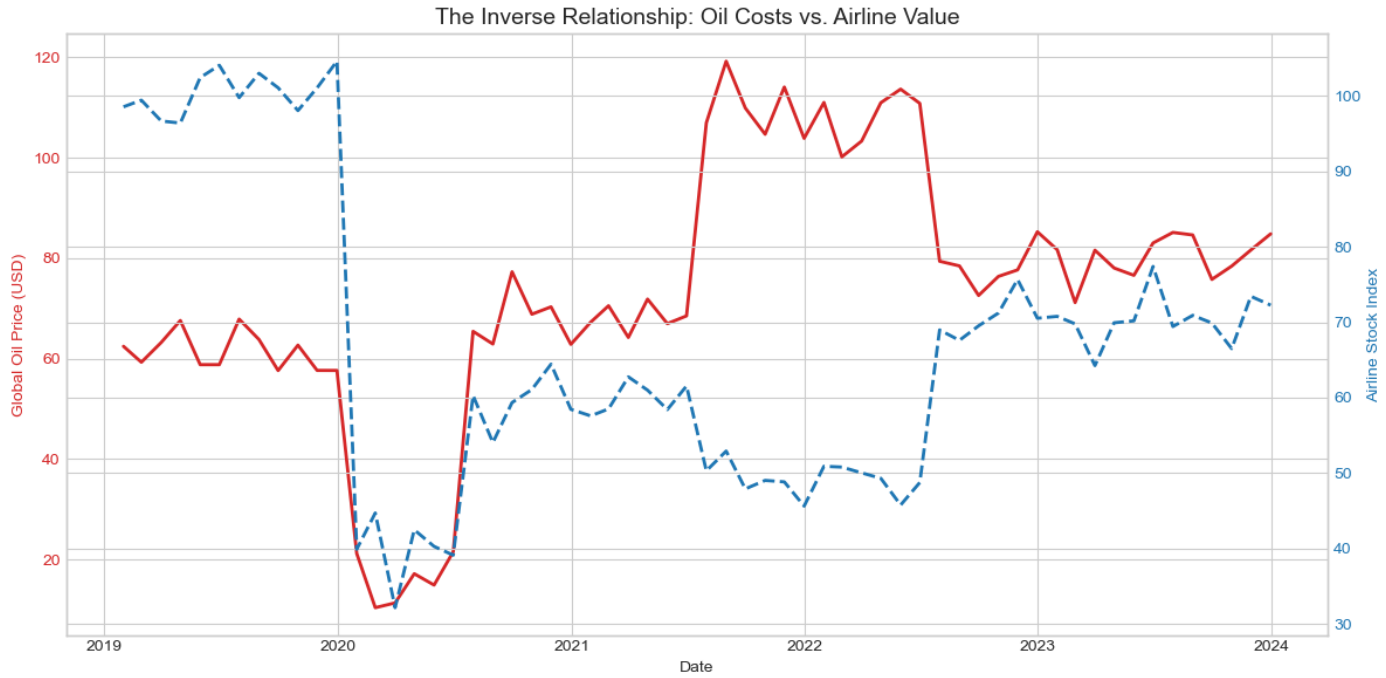
Simulating Shocks with Widgets

- We use ipywidgets to create sliders.
- **Task:** You will play "God" with the economy.
  - Create a Demand Shock (Consumer Panic).
  - Create a Supply Shock (Disaster).
- **Observation:** Watch how the intersection point moves. Does it match theory?



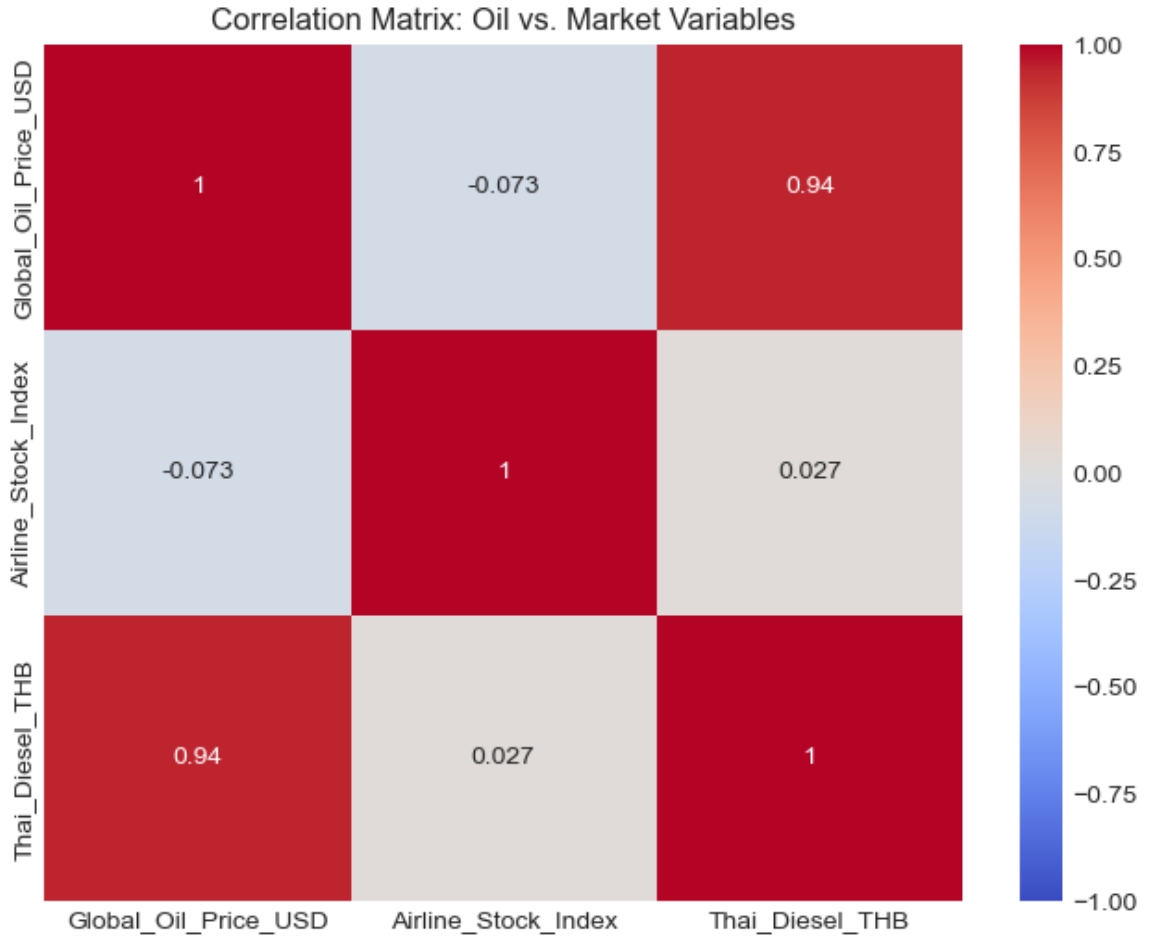
Analyzing Real Oil Data

- **Data Points:**
  - Global Oil Price (USD).
  - Airline Stock Index.
  - Thai Retail Diesel Price.
- **Visual:** We will plot these on dual axes to see the relationships.



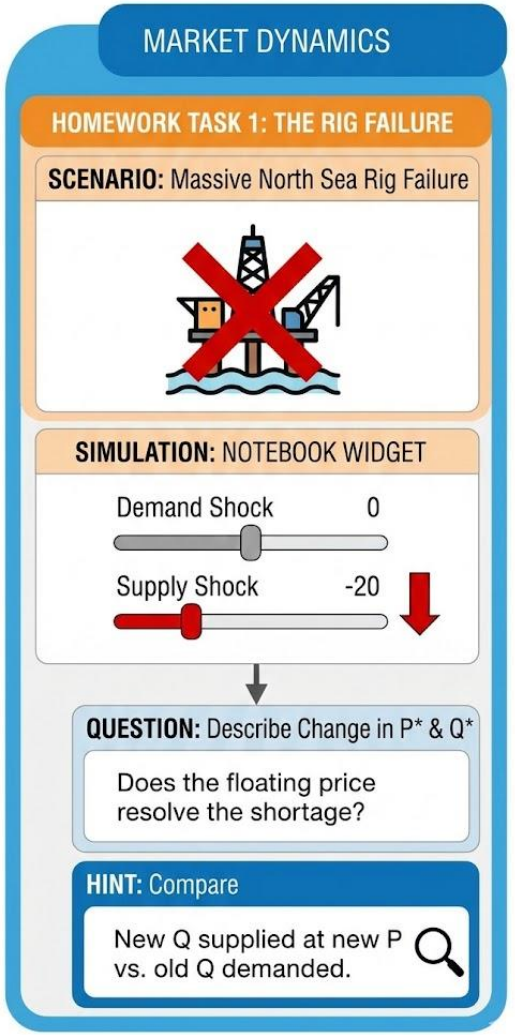
Correlation Heatmap

- We use sns.heatmap to quantify relationships.
- **Interpretation:**
  - **+1 (Red):** Perfect Positive Correlation (Oil & Diesel).
  - **-1 (Blue):** Perfect Negative Correlation (Oil & Airline Stocks).



Homework Task 1: The Rig Failure

- **Scenario:** A massive failure in North Sea oil rigs.
- **Simulation:** Use the notebook widget.
  - Set Demand Shock = 0.
  - Set Supply Shock = -20.
- **Question:** Describe the change in P^* and Q^*. Does the floating price resolve the shortage?
- **Hint:** Compare the *new* quantity supplied at the *new* price vs the *old* quantity demanded.



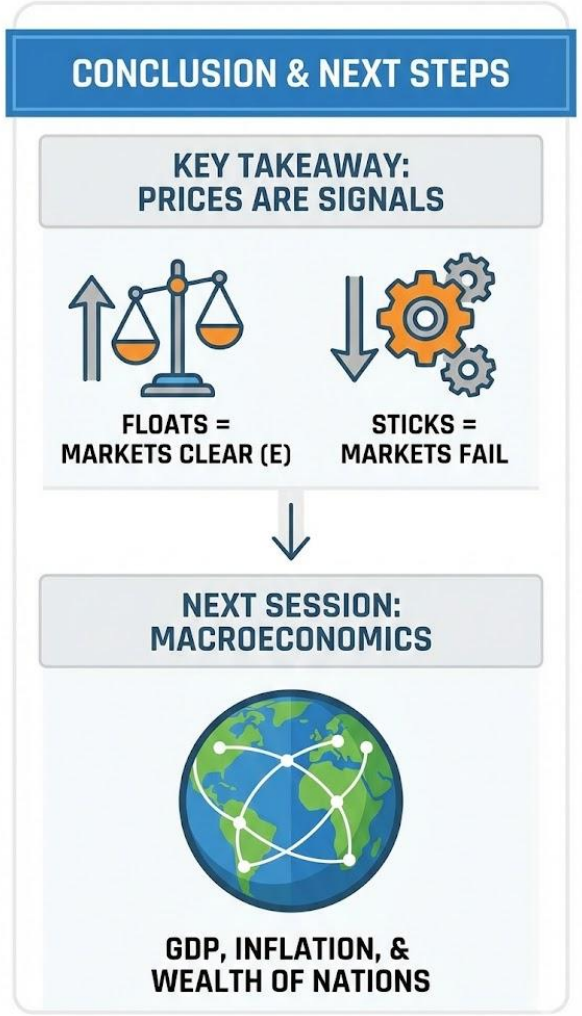
Homework Task 2: The Negative Price Anomaly

- **Scenario:** April 2020. WTI Crude hits -$37.
- **Task:** Use Supply/Demand logic to explain this.
- **Keywords to use:** "Demand Shock," "Storage Capacity," "Cost of Carry."
- **Critical Thinking:** Why did producers pay people to take the oil? Why couldn't they just pour it out? (Environmental laws/Shutdown costs).



Conclusion & Next Steps

- **Today:** We mastered the Market Mechanism (D+S=E).
- **Key Takeaway:** Prices are signals. When they float, markets clear. When they stick, markets fail.
- **Next Session:** We dive into **Macroeconomics**.
  - **Session 5:** GDP, Inflation, and the Wealth of Nations.



Let’s check out our Notebook

